THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which pair of quantities is represented by the symbols m and v in science? 1

(a) metre and volume **(b)** moles and voltage
(c) mirror and vision **(d)** mass and velocity

2. Climate change and designing sustainable technology need ideas from: 1

(a) several branches of science together **(b)** no science
(c) only physics **(d)** only biology

3. Estimating the rice needed for a family of four for a month: the answer 100 g is unreasonable because: 1

(a) it is exactly right **(b)** it is far too little for a whole month
(c) rice cannot be weighed **(d)** estimates must be exact

4. In earth science, the Earth is sometimes modelled as: 1

(a) a single point
(b) a smooth sphere layered into distinct regions
(c) a hollow cube
(d) a flat disc

5. Measurements in science are based on agreed international standards rather than: 1

(a) units **(b)** instruments
(c) local objects or opinions **(d)** numbers

6. Fog is a mixture of: 1

(a) solid dust in water **(b)** tiny liquid droplets in air
(c) gas in a solid **(d)** two solids

7. Muddy water is an example of a: 1

(a) true solution **(b)** suspension **(c)** pure substance **(d)** colloid

8. Ethanol and water are miscible. Their mixture is: 1

(a) homogeneous **(b)** a suspension **(c)** heterogeneous **(d)** a colloid

9. The Tyndall effect would be seen MOST clearly in: 1

(a) clear water **(b)** a beam of light passing through fog
(c) pure alcohol **(d)** sugar solution

10. The size of the particles in a suspension is: 1

(a) equal to atoms **(b)** smaller than 1 nm

(c) too small to filter **(d)** large enough to be seen and to settle

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does mathematics help in science? 2

12. Give one example where an approximate answer is enough and one where an exact value is needed. 2

13. What is an alloy? Give one example. 2

14. Give one example each of a heterogeneous mixture where a gas is one of the components. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. State three habits of thinking that the study of secondary science develops. 3

16. A mixture contains sand, salt and iron filings. Suggest the order of steps to separate all three. 3

17. Why must scientific terms be used ‘unambiguously’? Illustrate with one term. 3

OR

Why are measurements based on agreed international standards and not on local objects or opinions? Give one example from daily life. 3

18. The mixture of an immiscible liquid pair of the same density forms no layers. Explain. 3

OR

Why are most gas–gas mixtures homogeneous? Give one example and one example of a heterogeneous mixture of a gas. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State the steps of a good scientific approach to a numerical problem. 5
(ii) Apply them to estimate the time to read a 200-page book.
(iii) Explain how to check your answer.

OR

(i) Give one example of a model each in physics, chemistry, biology and earth science. 5
(ii) State what each model ignores.
(iii) Why is it acceptable to use different models for the same object?

20. (i) State the difference between evaporation, crystallization and distillation. 5
(ii) For each, name a mixture that can be separated by it.

OR

(i) What are the different types of heterogeneous mixtures? 5
(ii) Give two examples each of mixtures with a gas, a liquid and a solid as the dispersed phase.
(iii) Which of them show the Tyndall effect?

END OF QUESTION PAPER Total: 40 Marks